DATA ITEM DESCRIPTION

Title: MODEL-BASED ENGINEERING FAILURE MODES, EFFECTS, AND CRITICALITY ANALYSIS PROFILE (SYSML VERSION)

Number: DI-SESS-82426
AMSC Number: N10444
DTIC Applicable: Yes
http://www.dtic.mil/dtic/submit
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20231030
Limitation: N/A
GIDEP Applicable: Yes
http://www.gidep.org/data/submit.htm
Project Number: SESS-2023-041

Use/relationship: The failure modes, effects, and criticality analysis (FMECA) is a foundational reliability and maintainability (R&M) engineering design activity implemented during the design and development of a product or system. The FMECA identifies potential failure modes for independent single-item or functional failures, as well as the resulting effects on mission success, performance, safety, reliability, and maintainability. The FMECA, when performed early in the design process, allows for the implementation of appropriate corrective actions needed to eliminate Severity I and II failure modes or control risks associated with each remaining failure mode. Controlling critical failures can improve operational readiness and reduce life cycle cost.

This Systems Modeling Language (SysML) profile supports the move from static document delivery in this DID to interactive model delivery of FMECA data to allow FMECA data to be dynamically provided to the program office and other users allowing quicker engagement, decreasing design cycle times, and increasing design influence. This enables reliability and other engineering functions to continuously exchange information to identify and eliminate or control failure modes throughout a system's life cycle and apply the information as a basis for detecting and isolating faults, determining operator and maintainer actions, and identifying test requirements. The MBE approach allows engineers to interrogate the design to address the design views in Table 1. Figure 1 is an example implementation using a SysML profile.

This profile contains standard data elements needed to generate MBE FMECA design views as required by the specific and discrete tasks described in the solicitation (contract statement of work). The selection of required design views should be tailored appropriately by selecting only those views in Table 1 needed for the program.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described in the contract SOW.

To be considered compliant, with this DID, it is encouraged to use SysML to add and manage the data elements listed in this DID and report the design views with the properties described in this DID. This does not mean that it is required to use the accompanying MBE FMECA Profile; however, it can be used as an out-of-the-box tool to allow a team to easily be compliant with this DID.

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This DID is intended to prescribe the FMECA data elements that need to be captured for a system and define the Design Views, or the format that the data should be reported back in. The SysML Profile is intended to be an out of the box SysML tool to assist in this DID compliance by:

- Allowing the capture of the prescribed data elements through the use of a FMECA stereotype and tags
- Allowing the compilation of those data elements into the required design views by using generic tables

While the provided MBSE FMECA Profile will assist in compliance, with this DID, it is not required to be used. The MBSE Team is welcome to modify the provided FMECA Profile or create their own, provided that it captures the DID Data Elements and Design Views.

Further instructions for how to implement, model, and customize the accompanying SysML Profile can be found in the model itself.

Requirements:

1. Reference Documents.

1.1. Integrated Process. The FMECA process shall be integrated within the digital system model. The Digital System Model shall represent the functions described in the system specification requirements as elements of the model.

1.2. Model Interrogation. The MBE approach shall provide methods and access for the government to interrogate the digital system model to view data elements and their relationships per the pre-defined design views in Table 1. The interrogation shall have the ability to query, filter, or sort by:

1.2.1. Data Element e.g., *A1, A4, B2, F3*.

1.2.2. Design View or Design View Subcategory e.g., *1.3, 2.1, 2.3, 6.4*.

1.2.3. Metachain (Combination of data elements not defined in Table 1 Design Views).

2. Format. The FMECA shall be delivered in the native file format(s) of the modeling tool(s) or developed directly in the Integrated Digital Data Environment (IDDE) as identified in the Contractor's Systems Engineering Management Plan. The FMECA shall include documentation regarding necessary software tool(s) configurations, access types (e.g., readonly or write access), dashboards, views, data integrations, data flows, reporting mechanisms outside of dashboards (e.g., report building tools), security, size, and configuration management details.

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3. Content. The design views and associated data elements provided in Table 1, document the relationship of the FMECA to the SysML model, digital engineering artifacts, and R&M engineering analyses, as required by the contract statement of work.

3.1. The FMECA shall state how it is incorporated into the Digital System Model.
3.2. The FMECA shall state how flows are analyzed.

3.3. Functional Traceability. The Digital System Model shall contain a traceability matrix demonstrating that all functions requested in the specification requirements are covered.

3.4. Flows. The FMECA data shall be linked to model elements of the Digital System Model representing power flows between LRUs; fluid flows (e.g., fuel, cooling) between LRUs; internal communications flows between software components; external communications flows between the system and other systems or actors in the system context; and other signal flows such as relay closures, synchronization clock signals, and similar flows between line replaceable units and/or software components.

Table 1 – Design Views

No.	View Name	Mapped Data Elements (Columns)
1	Failure modes and effects summary (FMES)	A1, A2, B1, B5, C2, D2, D3, D4, E1, E2, F1, F2, F3, G1, G2, I1
2	Failures (based on the method of indenture; hardware, software, or functional)	A1, B1, B2, B3, C1, C2, C3, G1, J1, J2, J4
3	Diagnostics	A1, B1, C1, C2, C3
4	Hidden failures	A1, B1, B5, D2, D3, D5, G1
5	Compensating provisions	A1, B1, D2, D3, G1, G3
6	Actions taken	A1, B1, K1, K2, K3, K4, K5, K6

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7	Failure modes for which PM may be applicable	A1, B1, D2, H2
8	System Effects	A1, A2, B1, C2, D2, D3, F1, G1
9	Failure Propagation of each failure mode	A1, A2, B1, B2, C2, D1, E1, F1
10	Criticality Analysis Table	A1, A2, A3, A4, B1, B2, B3, B4, G1, G2, J1, J2, J3, J4, J5, J6, J7, J8
11	FMEA Table	A1, A2, A3, A4, B1, B2, B3, B4, B5, C1, C2, C3, D1, D2, D3, D4, E1, E3, F1, F3, G1, G2, H1, H2, I1

Note: All columns must be filterable and sortable

3.5. Data elements.

3.5.1. Item Function.

3.5.1.1. A1 – Identification Number: Serial number, Logistics Control Number (LCN) or other reference designation identification number is assigned for traceability purposes.

3.5.1.2. A2 – Item Identification Nomenclature: The name or nomenclature of the item or system function being analyzed for failure mode and effects.

3.5.1.3. A3 – Function: A concise statement of the performance required by the function of the item including any performance standards.

3.5.1.4. A4 – Function Failure Description: Ways an item can either partially (with upper and/or lower performance limits) or totally fail to perform its functions.

3.5.1.4.1. Incorrect Disposition: The item is not in a state that bears the desired function when the function is called.

3.5.1.4.2. Interrupted Change of Disposition: The item exited from state such that it can no longer physically complete the desired function.

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3.5.1.4.3. Performance: The item was in and remained in proper state and physically performed the desired function, but some quality of the function's execution did not meet performance.

3.5.2. Failure Modes and Effects.

3.5.2.1. B1 – Failure Modes: From the data entity perspective, the description of the relational difference between the desired condition and the predicted failed condition pointing from desired to fail. For incorrect disposition type failures, this will point between two operational states. For interrupted disposition type failures, this will point from operational to failed states. For performance failures, this will point from predicted (failed) performance to performance requirement (e.g., open, short, low torque, loss of power, etc.).

3.5.2.2. B2 – Failure Causes: Identification and description of the root cause(s) of the failure modes. The model should include process models that lead to a failure mode. Root cause is the point of deviation that leads to the failure mode (e.g., failure mechanism such as fracture, fatigue, corrosion, spalling, etc.).

3.5.2.3. B3 – Operational/Mission Phase: Description of the required higher-level process within which the failure may occur. Performed for all mission profile(s), under worst-case conditions.

3.5.2.4. B4 – Operational Mode/State: Description of the operational state of an item defined by a set of functions.

3.5.2.5. B5 – Failure Effects: Performance effects that identify the impact of a failure mode on the operation and function of a linked item.

3.5.3. Failure Detection and Isolation.

3.5.3.1. C1 – Failure Signatures: Unique set of characteristics that will support detection and isolation of the failure.

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3.5.3.2. C2 – Failure Detection Method: Description of the methods by which occurrence of the failure mode is detected by the operator or maintainer at the system level.

3.5.3.3. C3 – Isolation Procedure: Description of the most direct procedure that allows a maintainer to isolate the malfunction or failure. Describes the method by which ambiguities are resolved when more than one failure mode causes the same failure indication. Identifies to what indenture level failures can be isolated by the use of BIT features and indicate when ancillary test equipment will be required for failure isolation.

3.5.4. Compensating Provisions.

3.5.4.1. D1 – Compensating Provisions (Design / Operator): Compensating provisions, either design provisions or operator actions, that circumvent or mitigate the effect of the failure.

3.5.4.2. D2 – Protection: Protective or preventative measures to prevent the failure effect from occurring.

3.5.4.3. D3 – Mitigation: Mitigation of the end effect.

3.5.4.4. D4 – Protective Comments: Explanation of the protective measure or documentation of uncertainty.

3.5.4.5. D5 – Implemented By: The system feature, component, or instructions that allow the operator to implement the compensating provisions.

3.5.5. Intermediate Interactions.

3.5.5.1. E1 – Intermediate Effects: Identification of each of the effects (secondary failures) as the primary failure propagates through the system until its end effect.

3.5.5.2. E2 – Intermediate Causes: The failure causes associated with each of the intermediate effects.

3.5.5.3. E3 – Intermediate Items: Identification of each item along the failure's path of propagation through the system until its end effect.

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3.5.5.4. E4 – Intermediate Detections: List of all detection mechanisms other than the primary and end effect detections along the propagation path.

3.5.5.5. E5 – Intermediate Protections: List of all protection measures other than the primary and end effect protection measures along the propagation path.

3.5.5.6. E6 – Intermediate Mitigations: List of all mitigations other than the primary and end effect mitigations along the propagation path.

3.5.5.7. E7 – Intermediate Comments: Explanation or uncertainties on the intermediate detections, mitigations, or protections.

3.5.6. End Effects.

3.5.6.1. F1 – End Effects: End effects used to evaluate and identify the total performance effect of a failure on the mission and the surrounding environment.

3.5.6.2. F2 – End Cause: The cause of the failure at the end item.

3.5.6.3. F3 – End Item: The item at which the failure propagation terminates.

3.5.7. Severity.

3.5.7.1. G1 – Severity Classification: Categories (I-IV) assigned to each failure mode, without considering compensating provisions, according to the failure effect at the uppermost system. The severity classification categories are:

3.5.7.1.1. Category I – Catastrophic: A failure that may cause death or system loss (i.e., aircraft, tank, missile, ship)

3.5.7.1.2. Category II – Critical: A failure that may cause severe injury, major property damage, or major system damage that will result in mission loss.

3.5.7.1.3. Category III – Marginal: A failure that may cause minor injury, minor property damage, or minor system

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damage that will result in delay or loss of availability or mission degradation.

3.5.7.1.4. Category IV – Minor: A failure not serious enough to cause injury, property damage, or system damage, but will result in unscheduled maintenance or repair.

3.5.7.2. G2 – Severity Comment: Explanation or uncertainty in determining the severity.

3.5.7.3. G3 – Unmitigated Severity Classification: Categories (I-IV) assigned to each failure mode, without the consideration of any compensating provisions, according to the failure effect at the uppermost system.

3.5.8. Maintenance Actions.

3.5.8.1. H1 – Basic Corrective Maintenance Actions: The basic actions taken by the maintainer to correct the failure (e.g., remove and replace LRU)

3.5.8.2. H2 – Basic Preventative Maintenance Actions: The basic actions taken by the maintainer to prevent the failure (e.g., oil change).

3.5.9. Design Remarks.

3.5.9.1. I1 – Remarks: Any pertinent remarks or contractor's rationale pertaining to and clarifying any other columns or recommendations for design improvements.

3.5.10. FMEA/FMECA Data.

3.5.10.1. J1 – Failure Probability: Assessment of failure modes in terms of probability of occurrence (Levels A-E):

3.5.10.1.1. Level A – Frequent: High probability of occurrence, which may be defined as a single failure mode probability greater than or equal to 10^{-1} of the overall probability of failure during the operating time interval.

3.5.10.1.2. Level B – Reasonably probable: Reasonable probability of occurrence, which may be defined as a

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single failure mode probability of occurrence that is more than 10^{-2} but less than 10^{-1} of the overall probability of failure during the item operating time.

3.5.10.1.3. Level C – Occasional: Occasional probability of occurrence, which may be defined as a single failure mode probability of occurrence that is more than 10^{-3} but less than 10^{-2} of the overall probability of failure during the item operating time.

3.5.10.1.4. Level D – Remote: Remote probability of occurrence, which may be defined as a single failure mode probability of occurrence that is more than 10^{-6} but less than 10^{-3} of the overall probability of failure during the item operating time.

3.5.10.1.5. Level E – Extremely unlikely: Extremely unlikely, which may be defined as a single failure mode probability of occurrence that is less than 10^{-6} of the overall probability of failure during the item operating time.

3.5.10.2. J2 – Failure Rate Data Source: When failure rate data are to be used in the calculation of criticality numbers (item W), the data source of the failure rates used in each calculation. When a failure probability is listed, the remaining items J3-J6 are not required.

3.5.10.3. J3 – Failure Effect Probability (β): The conditional probability that the failure effect will result in the identified criticality classification, given that the failure mode occurs. The values represent the analyst's judgment as to the conditional probability the loss will occur and should be quantified in accordance with the following:

3.5.10.3.1. Actual Loss: = 1.0

3.5.10.3.2. Probable Loss: > 0.10 to < 1.0

3.5.10.3.3. Possible Loss: > 0 to = 0.10

3.5.10.3.4. No Effect: = 0

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3.5.10.4. J4 – Failure Mode Ratio (α): The fraction of the part failure rate related to the particular failure mode under consideration.

3.5.10.5. J5 – Failure Rate (λ): The failure rate from the appropriate reliability prediction.

3.5.10.6. J6 – Operating Time (t): The operating time of the item per mission derived from the government-approved operational concept or System Requirements Specifications.

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3.5.10.7. J7 – Failure Mode Criticality Number [$c_m = (\beta \alpha \lambda t)$]: A relative measure of consequence of a failure mode and its frequency of occurrence.

3.5.10.8. J8 – Item Criticality Number [$c_r = \sum_{n=1}^{i}(\beta \alpha \lambda t)$]: A relative measure of consequence of an item failure and its frequency of occurrence.

3.5.11. Design Actions Taken.

3.5.11.1. K1 – FMEA Prevention Action: Description of how a potential cause resulting in a given failure mode is mitigated using existing and planned activities.

3.5.11.2. K2 – FMEA Detection Action: The simulations or tests available to detect that the potential failure mode during the manufacturing or production process.

3.5.11.3. K3 – Responsible Organization: Organization responsible for corrective action.

3.5.11.4. K4 – Target Completion Date: Date corrective action is due.

3.5.11.5. K5 – Status: Current status of corrective action development.

3.5.11.6. K6 – Actions Taken with Reference to Evidence: Description of the corrective action and evidence that it was effective.

3.5.12. Schema – Example of how to apply data element stereotypes to capture system

FMECA data (Figure 1)

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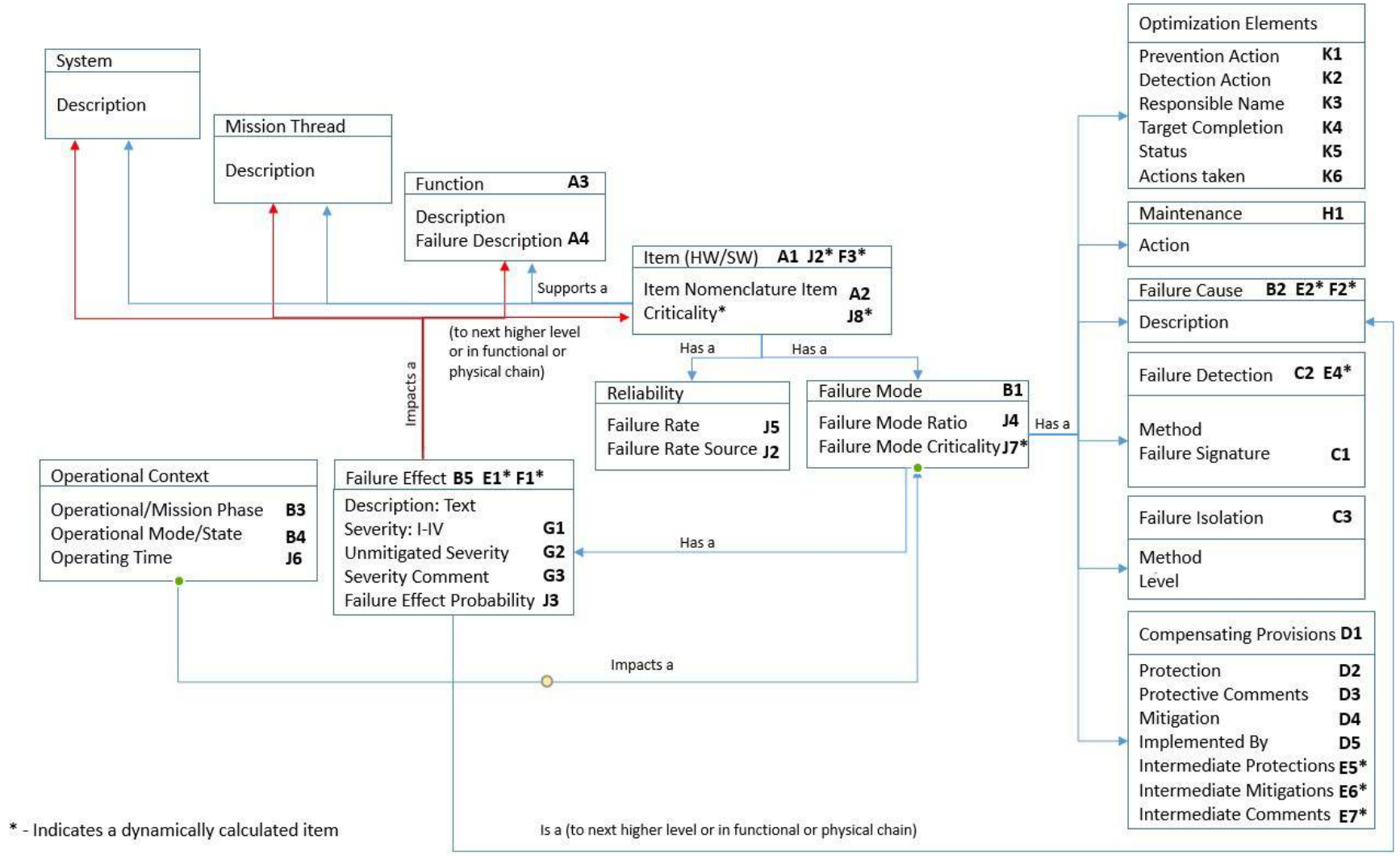


Figure 1: Example of Schema for Application of SysML Profile

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